

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CHEMICAL COORDINATION & INTEGRATION

LECTURE-03

ZOOLOGY

DAVOOD SIR (CEO of NG)



Physics Wallah

Topics To Be Covered

1

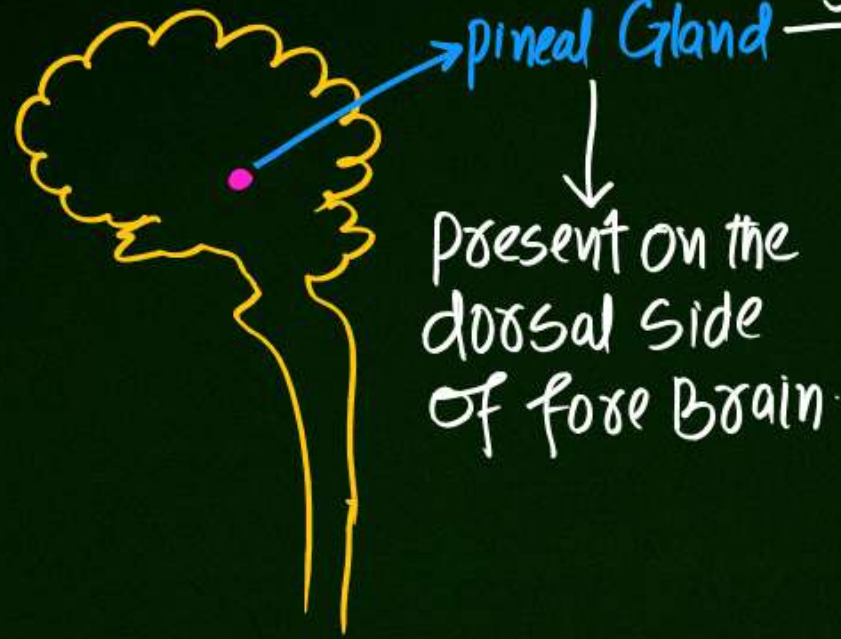
ENDOCRINE GLANDS.

- Pineal Gland ✓
- thyroid gland ✓
- Parathyroid gland ✓
- Thymus gland ✓
- Pancreas ✓

2

PREVIOUS & CURRENT YEAR NEET MCQ's.

Pineal Gland



pineal Gland

present on the dorsal side of fore Brain

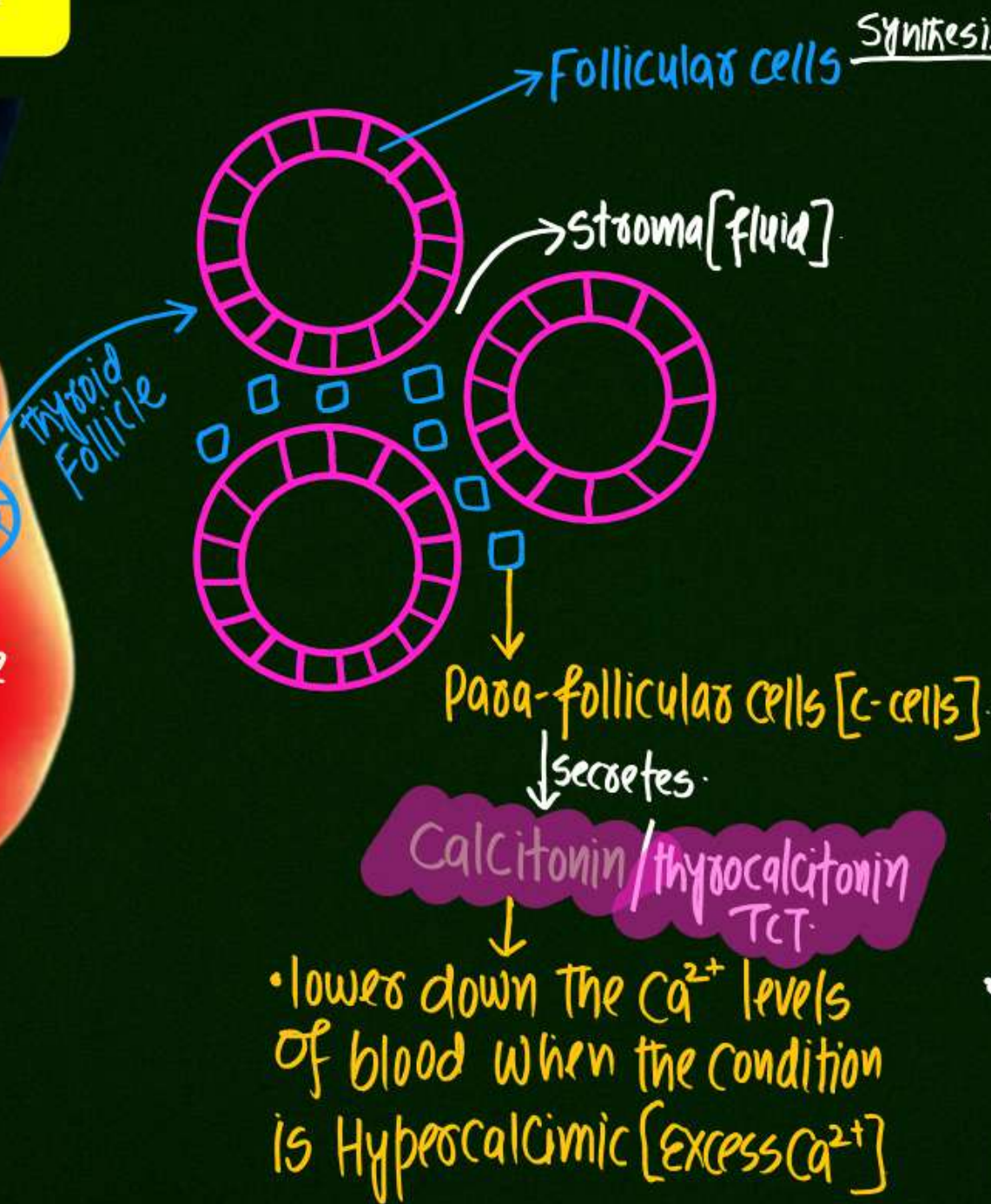
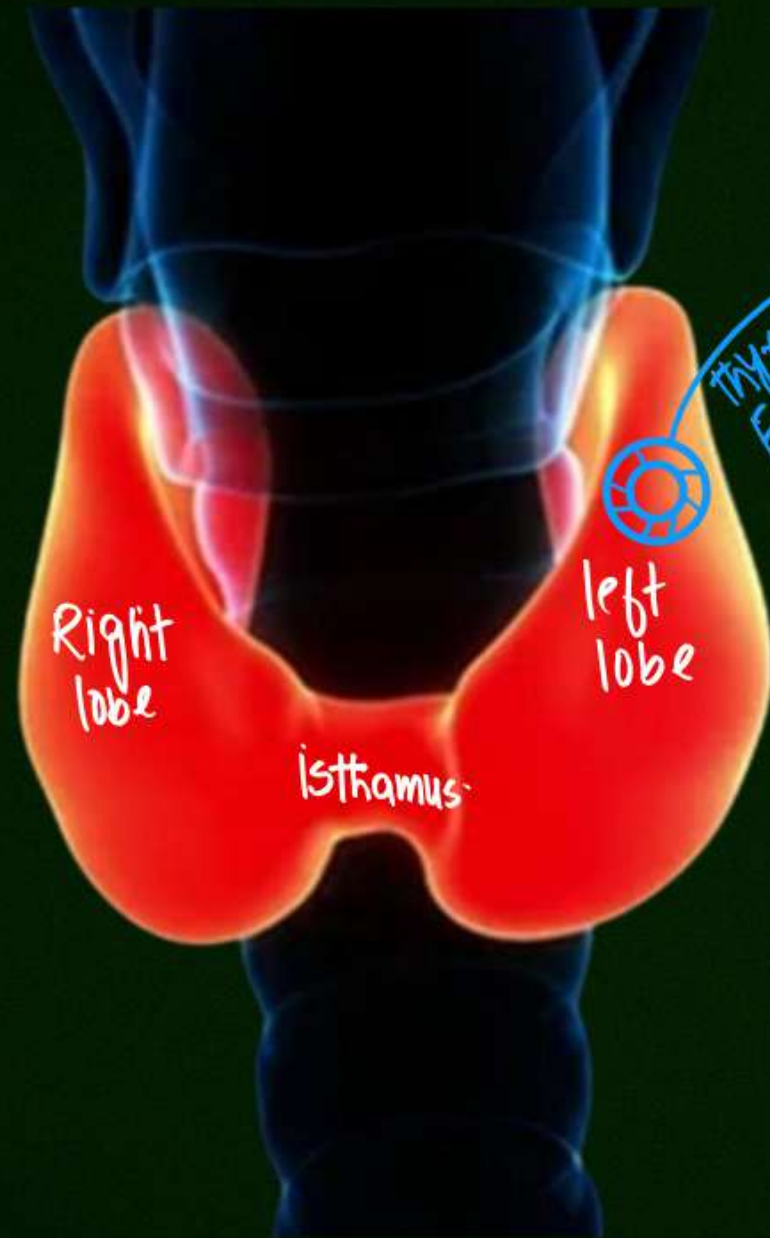
secretes

Melatonin Hormone

Functions

- Regulates diurnal rhythm (24hr) cycle or sleep wake cycle.
- Helps in pigmentation + Metabolism.
- Helps in Bodies defence capability.
- Helps in regulation of menstrual cycle.

Thyroid Gland



Synthesize

- Triiodothyronine [T_3]** → Made of 3-Iodine
- Tetraiodothyronine [T_4]** → Made of 4-iodine
Thyroxine

- Deficiency of Iodine leads to Hypothyroidism and causes enlargement of thyroid gland called Goitre.
- In pregnancy hypothyroidism causes stunted growth (cretinism), Mental Retardation, low IQ, Abnormal skin, deaf mutism in baby.
- In adult females, Hypothyroidism causes irregular menstrual cycle.
- Excess thyroid hormone production is called Hyperthyroidism which is due to cancer/nodules which cause increased BMR, weight loss, and disturbed body physiology.
- Exophthalmic Goitre (Graves disease) is thyroid enlargement with protruding eyeballs + weight loss.

Thyroid Gland

Functions of T_3/T_4 hormones

- (i) Regulation of Basal Metabolic Rate [BMR] and metabolism of Carbohydrate, protein etc.
- (ii) Influence electrolyte + H_2O balance.
- (iii) Helps in RBC formation.

Thyroid Gland

Goitre [thyroid gland enlargement]



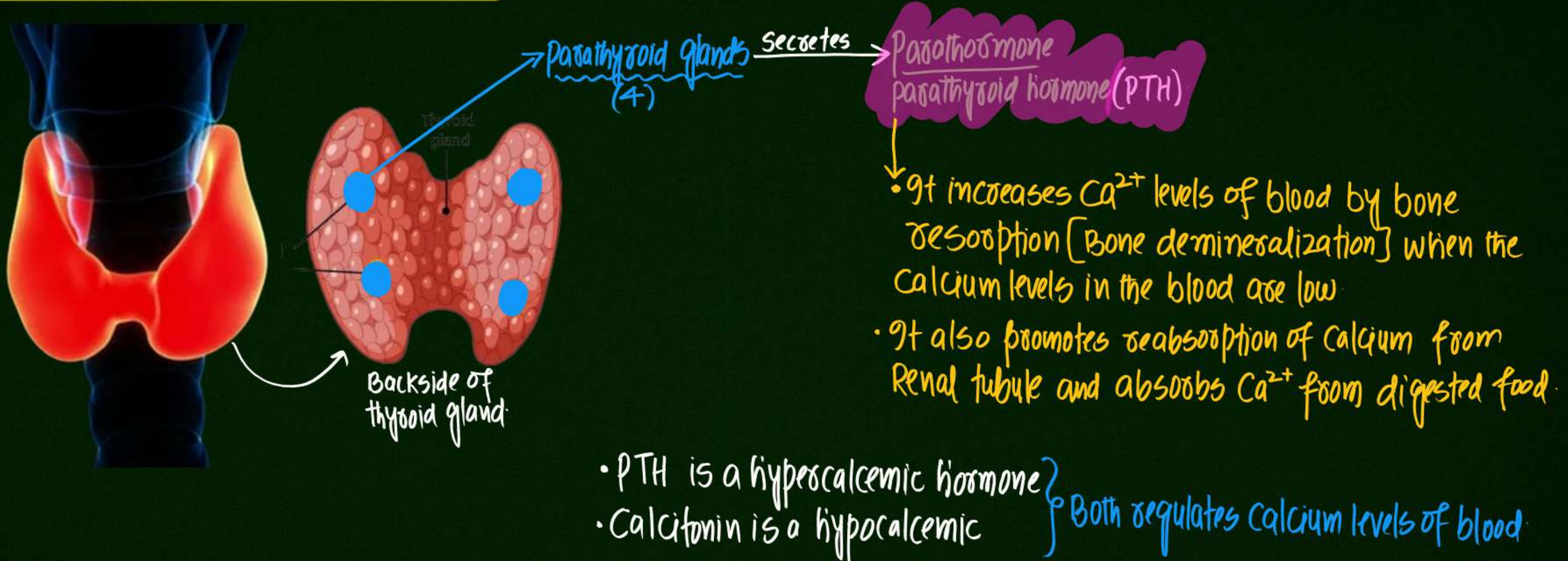
Cretinism (stunted growth)



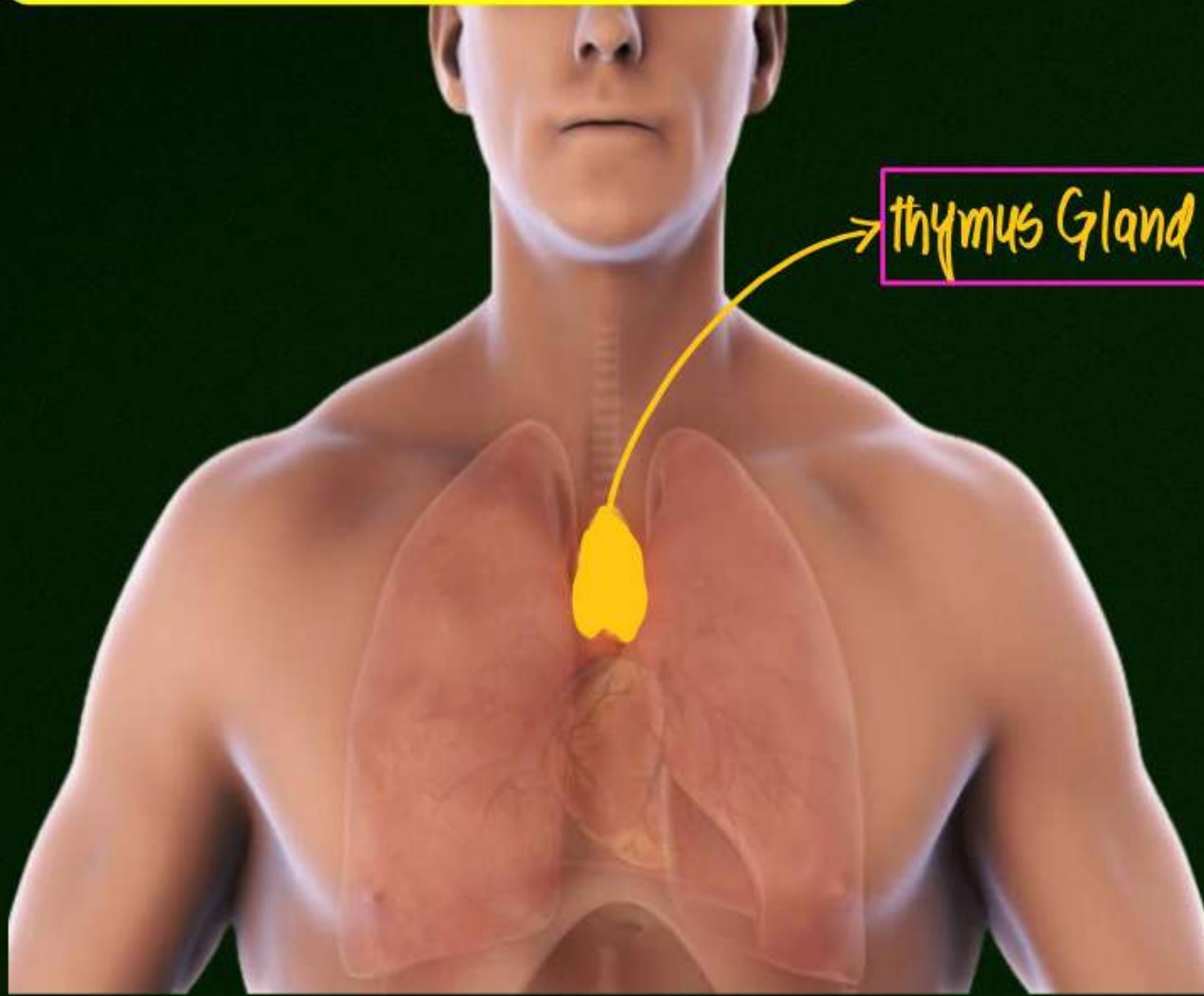
Exophthalmic Goitre [Graves disease]



Parathyroid Gland



Thymus Gland



thymus Gland

secretes

Thymosins

located in chest region behind sternum but ventrally to heart.

helps in Differentiation or maturation of T-lymphocytes. So, cell mediated immunity is created.

It also promotes antibody production & provides humoral immunity.

• in old age, the thymus gland degenerates → decreased thymosin production → Weak immunity → Disease.

Thank
You